

The quantitative relationship between government expenditure on tertiary education and income inequality

Koen Stokvis – 2859043 Oscar de Keijser – 2862760 Zoe Cheung – 2802886
Khumo Nasipkulov – 2908830 Majd al-Jawald – 2806525 Julia Smit – 2897717
Chantal Schouwenaar Tutorial 1 Group 2

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Part 1 - Identify a Social Problem

1.1 Summary of the Social Problem

Income inequality has been a prominent issue worldwide for many decades. Although some inequality is inevitable in a market-based economic system as a result of differences in talent, effort, and luck, excessive inequality can reduce economic growth by lowering consumer demand, increasing social tensions, and limiting opportunities for lower-income groups (International Monetary Fund, nd). In fact, recent empirical studies show that in many advanced economies, including Germany, Japan, and the United States, inequality has reached levels that negatively affect economic performance (Peterson & Schoof, 2015).

A possible approach to reducing income inequality is through increased investment in higher education. Education is widely seen as a fundamental driver of economic growth and welfare. Studies consistently show that higher levels of education are associated with greater individual earnings, lower poverty rates, and stronger long-term economic growth (Grant, 2017). Improving equal access to education could therefore promote both social equity and sustainable economic growth (Cingano, 2014).

In this report we quantify the relationship between income inequality and government expenditure on higher levels of education. We specifically examine the Gini-coefficient in European countries over the past years, compared to the share of government expenditure allocated to tertiary education.

Part 2 - Data Sourcing

2.1 Loading in the data

We downloaded three datasets as a csv.file from Eurostat (2026): ‘Total Government Expenditure’, ‘General Income Inequality’ and ‘Tertiary Education Expenditure’.

2.2 A short summary of the datasets

Dataset: Gini coefficient of equivalised disposable income

~Source of data: Eurostat ~Statistical theme: Tables on EU policy ; Population and social conditions ~Last data update: 08/06/2026 23:00 ~Data coverage: 2014 - 2025 ~Number of cells: 435

Dataset: Public educational expenditure by education level, programme orientation, type of source and expenditure category

~Source of data: Eurostat ~Statistical theme: Population and social conditions ~Last data update: 08/05/2026 23:00 ~Data coverage: 2012 - 2023 ~Number of cells: 634 963

Dataset: General government expenditure by function (COFOG)

~Source of data: Eurostat ~Statistical theme: Economy and finance ~Last data update: 27/04/2026 11:00 ~Data coverage: 1990 - 2025 ~Number of cells: 21 424 873

2.3 Summary of the types of variables used

The variables we have used for our project, from the raw data sets, contain: years, Gini coefficients and government spending in euro's (total and on tertiary education). We ourselves have made variables that are percentages and averages (see 3.2) (Eurostat, 2026).

The data from Eurostat came from administrative sources as well as questionnaires and surveys (Eurostat, 2026).

Part 3 - Quantifying

3.1 Data cleaning

We started the data cleaning process by selecting the variables needed for our research. These variables included years, observation values and countries. Next, We renamed some variables in the columns of every dataset to make programming with them easier, and for clarity. "OBS_VALUE" was initially found in multiple datasets but had different meanings, so we renamed it to something that better reflected the value represented in each dataset. For example in the tertiary education dataset "OBS_VALUE" was turned into "edu_exp". We also removed rows containing averages of multiple European countries across multiple years, which we did not need. Finally, we removed Kosovo because this country only had one year of data.

3.2 Generating the necessary variables used

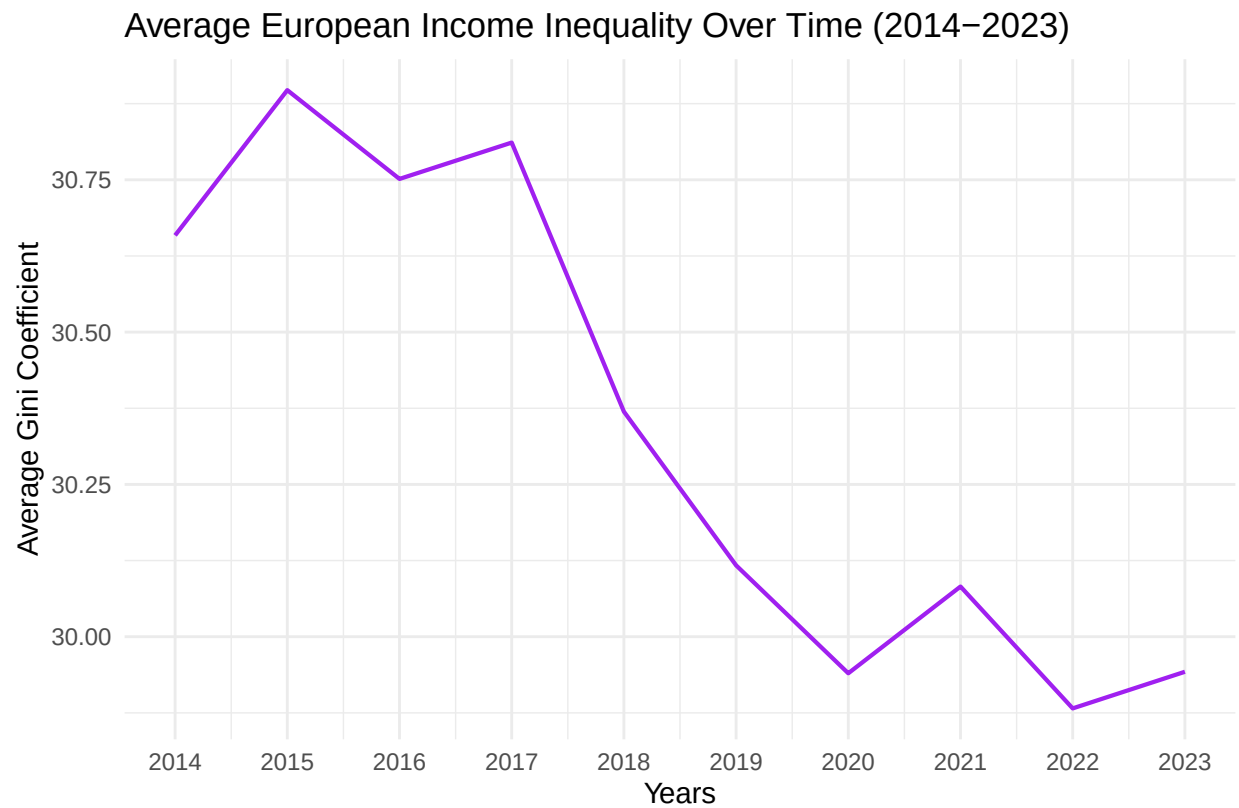
For our analysis we created several new variables to facilitate a better understanding of how expenditure on tertiary education affects the income inequality in Europe. First, we merged the tertiary education expenditure dataset with the total government expenditure dataset. Based on this merged dataset we created a new variable by dividing the tertiary education expenditure with the total government expenditure. This variable gives a more accurate representation of the tertiary education expenses by a country.

For our second variable we created the average Gini coefficient for each country by taking the mean of the Gini coefficients over the years (2014-2023). The same was done for the tertiary education variable for each country.

For our third variable we divided the countries within the datasets of tertiary education and general income inequality into four regions: 'North Europe', 'West Europe', 'South Europe' and 'East Europe' (WorldAtlas, 2025).

For our fourth variable we created the average Gini coefficient and the average tertiary education variable for all European countries in the datasets over time (2014-2023)

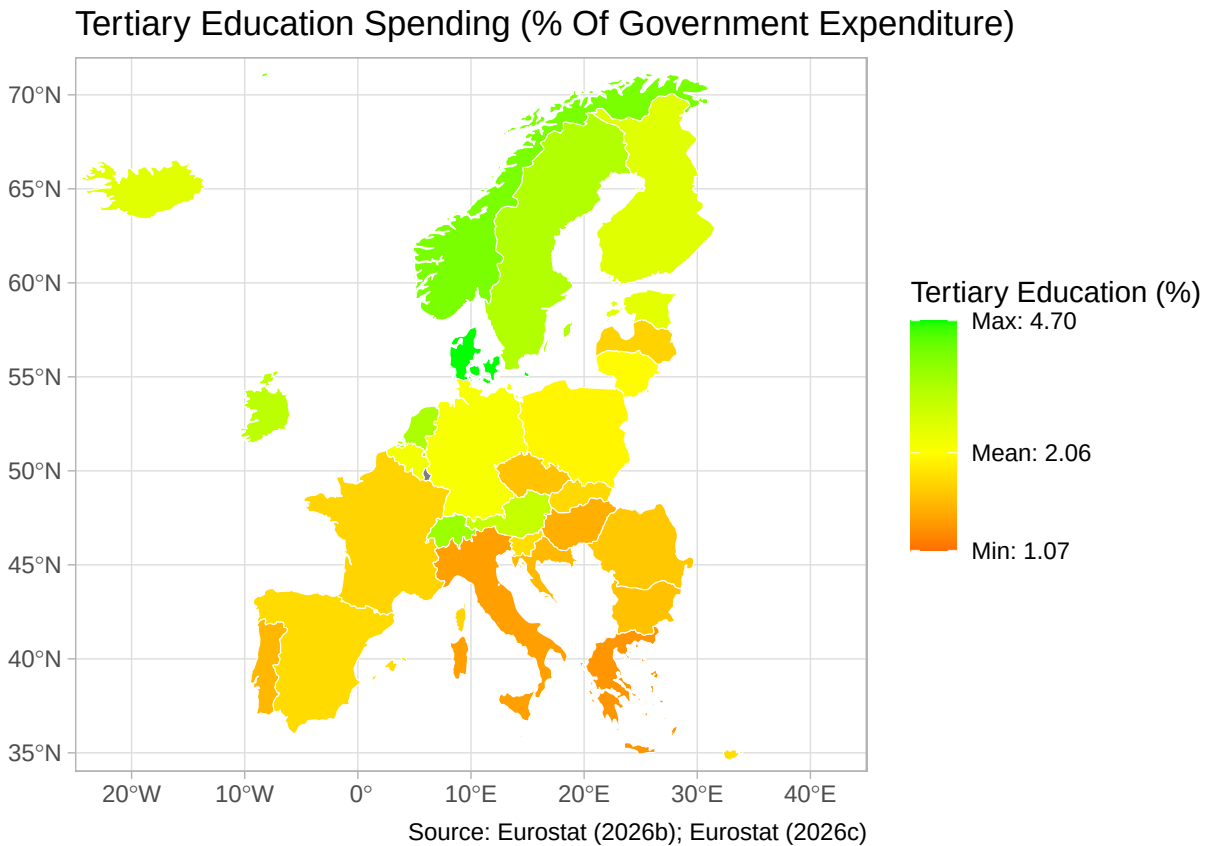
3.3 Visualizing temporal variation



Source: Eurostat (2026a)

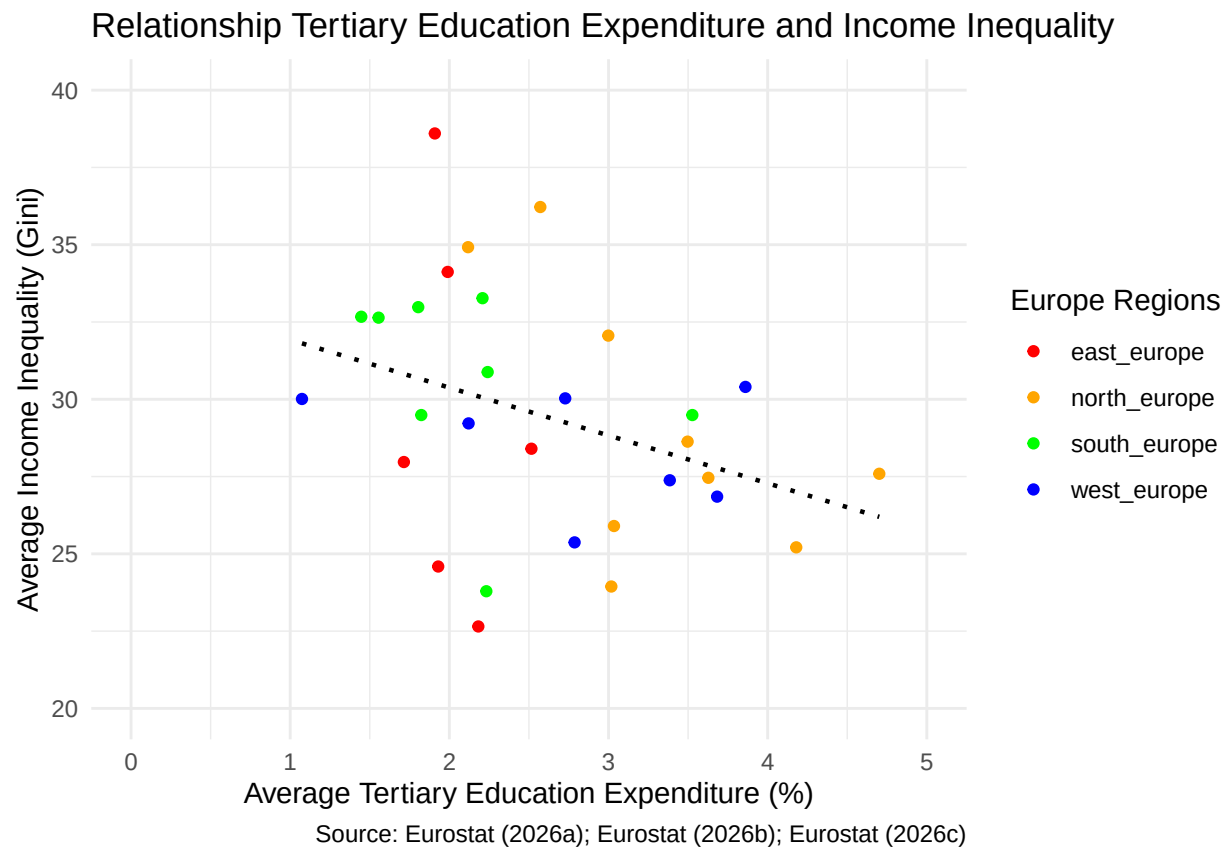
The average income inequality in Europe (see 3.2) is declining over a time period of nine years (2014-2019). With a significant decline from 2017 to 2020, falling back to a stable lower level from 2020 onward. It is important to note that the fluctuations across time are relatively small in magnitude, amounting to only a 1% change.

3.4 Visualizing spatial variation



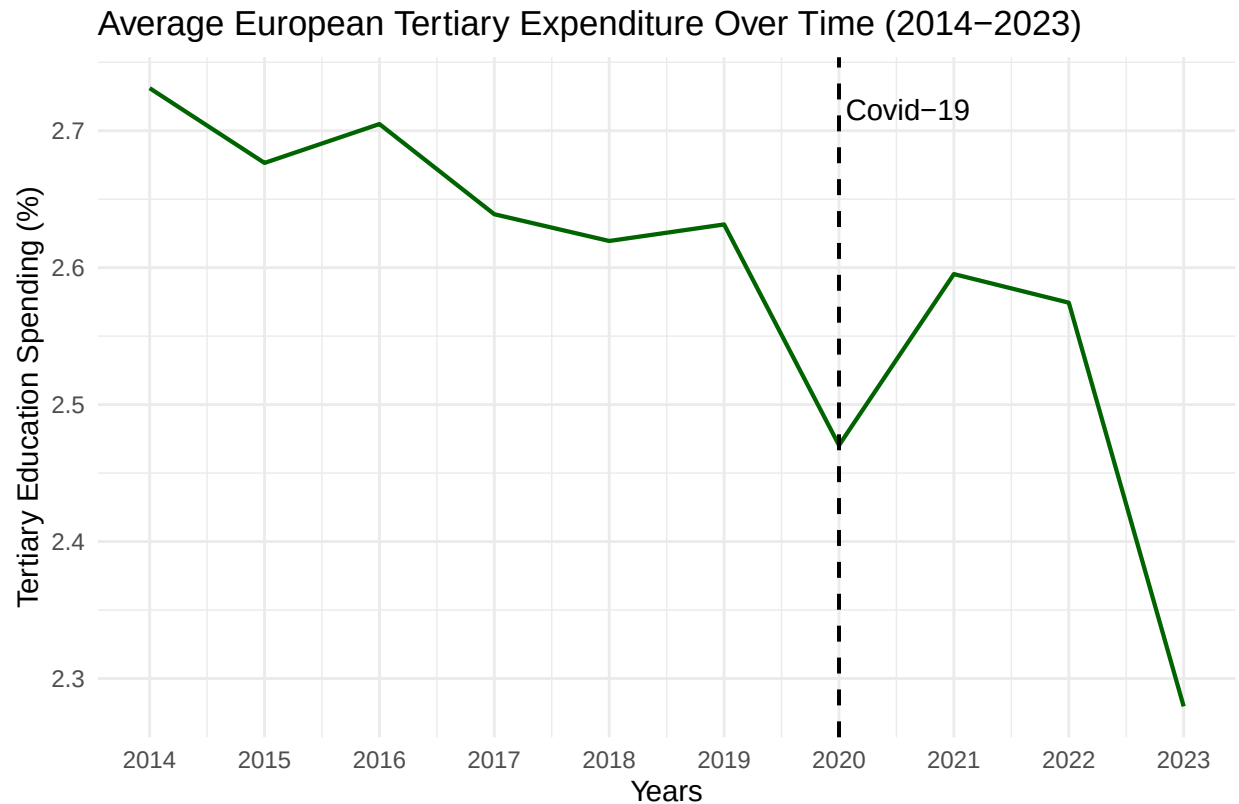
When looking at the spatial map of the tertiary education expenditure (see 3.2) a clear division between the spending of different countries can be seen. North and West Europe spend relatively more on tertiary education than South and East Europe. Denmark and Norway spend the most tertiary education, while Italy and Greece spend the least. The United Kingdom and Russia were not included in the dataset.

3.5 Visualizing sub-population variation



Across all datapoints on the scatterplot, a trend line can be drawn that shows a negative correlation between the average tertiary education expenditure and average income inequality (see 3.2). North Europe has a large spread in both inequality and spending, with spending being on the higher end of the total distribution. West Europe has similar inequality but large spread in spending. South Europe has moderate spread in both inequality and spending. East Europe has similar spending but high spread in inequality.

3.6 Visualizing event analysis



Source: Eurostat (2026b); Eurostat (2026c)

The average tertiary education expenditure in Europe (see 3.2) goes down over a period of nine years (2014-2019). We see a sharp decline in 2020, which coincides with the time of the Covid-19 pandemic. After 2020 the spending recovers only to fall significantly in 2023.

Part 4 - Discussion

4.1 Discussing our findings

Our analysis shows that there is a negative relationship between government expenditure on tertiary education and income inequality (see 3.5). We found that the Gini coefficient has declined over the years meaning that income inequality has improved (see 3.3). However, we found that the tertiary education expenditure, as a percentage of total government spending, has also declined (see 3.6). This is in contradiction with the negative relationship we found. In our event analysis we observed a sharp decline in government spending in 2020. This decline is likely the result of a shortage in funding caused by the Covid-19 pandemic (Al-Samarrai et al., 2020). However, we can not state this relationship with absolute certainty. Our spatial analysis graph shows a high level of tertiary education spending in Scandinavia (see 3.4). In these Scandinavian countries there is a low level of income inequality (see dataset: income_ineq). Further research could examine why this negative relationship is present in Scandinavia.

All data used in this research is obtained from Eurostat. Even though relying on only one data source ensures consistency, it may limit the reliability of the findings. Furthermore, this analysis only considers the proportion of total government spending spent on education and does not account for differences in absolute expenditure between countries. In addition, income inequality is influenced by many factors. This

research focusses exclusively on the possible effect of educational spending and does not consider any other factors. Therefore, the results of this research can be seen as indicating an association, rather than a causal relationship. Future research could include additional determinants of income inequality and examine changes in absolute spending.

Part 5 - Reproducibility

5.1 Github repository link

<https://github.com/Oscar-de-Keijser/Expenditure-on-education>

5.2 Reference list

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